

GARCH Models

Topics:

- Volatility: Introduction
- The ARCH and GARCH models
- Estimation of GARCH models
- Specification Testing in GARCH models
- Volatility forecasting

Volatility: Introduction

Half of the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2003 was awarded to Robert F. Engle III *“for methods of analyzing economic time series with time-varying volatility (ARCH)”*

Volatility: Introduction

The *volatility* of an investment is a measure of its *risk*. Usually defined as the variance of the returns on the investment.

Volatility is an important ingredient in:

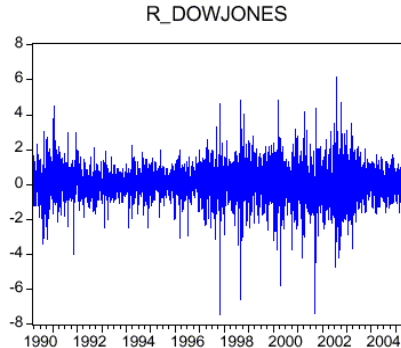
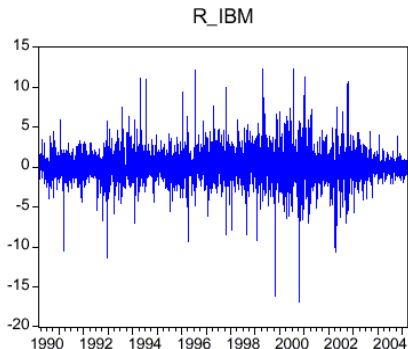
- portfolio selection;
- risk management;
- option pricing.

Empirical Evidence: Daily financial returns display *volatility clustering*: periods of high volatility alternate with more tranquil periods.

- This forms the basis for the *Autoregressive-Conditional Heteroskedasticity model* (Robert Engle) and the *Generalized Autoregressive-Conditional Heteroskedasticity* model (Tim Bollerslev).

Volatility: Introduction

Daily log-returns on IBM stock price and Dow Jones index, March 1990 – March 2005



The ARCH(1) model

The $AR(1)$ model $Y_t = c + \phi Y_{t-1} + \varepsilon_t$ (assuming that ε_t is a martingale difference sequence) can be formulated as $E_{t-1}(Y_t) = c + \phi Y_{t-1}$ where $E_{t-1}(\cdot) = E(\cdot | F_{t-1})$ where F_{t-1} is the information until and including period $t - 1$.

Assume for simplicity that assuming that $E_{t-1}(R_t) = 0$. In this case the first-order *autoregressive-conditional heteroskedasticity* (ARCH(1)) model for a return R_t is an $AR(1)$ for R_t^2 :

$$E_{t-1}(R_t^2) = \omega + \alpha R_{t-1}^2.$$

This model can be written as

$$\begin{aligned}\sigma_t^2 &= \text{var}_{t-1}(R_t) \\ &= E_{t-1}(R_t^2) \\ &= \omega + \alpha R_{t-1}^2.\end{aligned}$$

where $\text{var}_{t-1}(\cdot) = \text{var}(\cdot | F_{t-1})$. In practice we need to allow for $E_{t-1}(R_t) = \mu_t \neq 0$. Then $R_t = \mu_t + u_t$, $E_{t-1}(u_t) = 0$, and the model is $AR(1)$ for u_t^2 :

$$\sigma_t^2 = \text{var}_{t-1}(R_t) = E_{t-1}(u_t^2) = \omega + \alpha u_{t-1}^2.$$

Properties of ARCH(1)

It is possible to show that u_t is stationary if and only if $0 \leq \alpha < 1$. The variance is

$$\sigma^2 = \text{var}(u_t) = \frac{\omega}{1 - \alpha}$$

Defining $\eta_t = u_t^2 - \sigma_t^2$, the ARCH(1) model is an AR(1) model for u_t^2 :

$$u_t^2 = \omega + \alpha u_{t-1}^2 + \eta_t,$$

with $E_{t-1}(\eta_t) = 0$, hence $E(\eta_t) = 0$ and $\text{cov}(\eta_t, \eta_{t-l}) = 0$ for $l \geq 1$.

Assuming that $\text{var}(\eta_t)$ is finite u_t^2 is stationary.

The ARCH(q) model

When trying to estimate ARCH models one might find that more lags are needed, leading to ARCH(q):

$$\sigma_t^2 = \omega + \alpha_1 u_{t-1}^2 + \dots + \alpha_q u_{t-q}^2.$$

Note: Variances must be positive, a sufficient condition is $\omega > 0$, $\alpha_i \geq 0, i = 1, \dots, q$.

- A necessary and sufficient condition for stationarity of u_t is that $\sum_{i=1}^q \alpha_i < 1$ [and $\omega > 0, \alpha_i \geq 0, i = 1, \dots, q$].
- Corresponds to an AR(q) model for u_t^2 :

$$u_t^2 = \omega + \alpha_1 u_{t-1}^2 + \dots + \alpha_q u_{t-q}^2 + \eta_t,$$

with $\eta_t = u_t^2 - \sigma_t^2$, $E_{t-1}(\eta_t) = 0$, hence $E(\eta_t) = 0$ and $cov(\eta_t, \eta_{t-l}) = 0$ for $l \geq 1$ and assuming that $var(\eta_t)$ is finite.

- Note that the roots of $A(z) = 1 - \alpha_1 z - \alpha_2 z^2 - \dots - \alpha_q z^q$ must be outside the unit circle for u_t^2 to be stationary.
- If $\omega > 0, \alpha_i \geq 0, i = 1, \dots, q$ this is equivalent to $\sum_{i=1}^q \alpha_i < 1$.
- We can identify the model by checking the ACF and PACF of u_t^2 .

The GARCH(1,1) model

- A simpler structure than ARCH(q) is an ARMA(1,1) for R_t^2 or u_t^2 , which leads to the *generalized ARCH* model of orders (1,1) (GARCH(1,1)):

$$\sigma_t^2 = \omega + \alpha u_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \omega > 0, \alpha \geq 0, \beta \geq 0.$$

- For $\beta < 1$ this is equivalent to an ARCH(∞) model

$$\sigma_t^2 = \frac{\omega}{1 - \beta} + \sum_{j=0}^{\infty} \beta^j \alpha u_{t-1-j}^2.$$

- **Advantage:** We have less parameters to estimate.

The GARCH(1,1) model

- The GARCH(1,1) model is stationary if and only if $\alpha + \beta < 1$ and $\omega > 0, \alpha \geq 0, \beta \geq 0$. Thus

$$\sigma^2 = \frac{\omega}{1 - \alpha - \beta}.$$

- It corresponds to an ARMA(1,1) model for u_t^2 :

$$u_t^2 = \omega + (\alpha + \beta)u_{t-1}^2 + \eta_t - \beta\eta_{t-1}$$

with $\eta_t = u_t^2 - \sigma_t^2$. $E_{t-1}(\eta_t) = 0$, hence $E(\eta_t) = 0$ and $cov(\eta_t, \eta_{t-l}) = 0$ for $l \geq 1$. We assume that $var(\eta_t)$ is finite.

- The ACF and PACF of u_t^2 in case of stationary GARCH(1,1) are both exponentially decaying, no cut-off point;

GARCH(q,s) and IGARCH

Further generalisation GARCH (q,s)

$$\sigma_t^2 = \omega + \sum_{\ell=1}^q \alpha_{\ell} u_{t-\ell}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2, \omega > 0, \alpha_{\ell} \geq 0, \beta_j \geq 0$$

Remarks:

- Corresponds to $ARMA(p,s)$ for u_t^2 with $p = \max\{q,s\}$
- Stationary if and only if $\sum_{\ell=1}^q \alpha_{\ell} + \sum_{j=1}^s \beta_j < 1$ [and $\omega > 0, \alpha_{\ell} \geq 0, \beta_j \geq 0$].
- The acf and pacf of u_t^2 in case of stationary GARCH(q,s) are both exponentially decaying, no cut-off point;
- $\omega > 0, \alpha_{\ell} \geq 0, \beta_j \geq 0$ sufficient (not necessary) for $\sigma_t^2 > 0$.
- Model with unit root ($\alpha_1 + \beta_1 = 1$ in case of GARCH(1,1)):
Integrated GARCH (IGARCH): infinite variance, no mean-reversion in volatility.

The GARCH model

Some other properties:

- Let $\sigma_t = \sqrt{\sigma_t^2}$. The *standardized returns*

$$\varepsilon_t = \frac{R_t - \mu_t}{\sigma_t} = \frac{R_t - E_{t-1}(R_t)}{\sqrt{\text{var}_{t-1}(R_t)}}$$

satisfy $E_{t-1}(\varepsilon_t) = 0$ and $\text{var}_{t-1}(\varepsilon_t) = 1$. Therefore the model may be formulated as

$$\begin{aligned} R_t &= \mu_t + u_t = \mu_t + \underbrace{\sigma_t \varepsilon_t}_{u_t} \\ \sigma_t^2 &= \omega + \alpha u_{t-1}^2 + \beta \sigma_{t-1}^2. \end{aligned}$$

- Often it is assumed that ε_t are independent and identically distributed as $N(0, 1)$.

Kurtosis of GARCH models

For any weakly stationary process x_t with constant $E(x_t^4)$ define kurtosis as

$$k_x = \frac{E[(x_t - E(x_t))^4]}{\text{var}(x_t)^2}.$$

Measures the fatness of the tails of a probability density function.

- If $x_t \sim N(\mu_x, \sigma_x^2)$, then $k_x = 3$. Excess Kurtosis $K_x = k_x - 3$.
- **Empirical evidence:** financial data often displays more kurtosis than that permitted under the assumption of normality (fat tails.)
- $u_t = \sigma_t \varepsilon_t$ with $E(\varepsilon_t) = 0$ $\text{var}(\varepsilon_t) = 1$ and assuming that ε_t are i.i.d.:

$$k_u = \frac{E[\sigma_t^4 \varepsilon_t^4]}{\text{var}(\sigma_t \varepsilon_t)^2} = \frac{E[\sigma_t^4]}{E[\sigma_t^2]^2} k_\varepsilon \geq k_\varepsilon.$$

because by Jensen inequality $E(h(X)) \geq h(E(X))$ for a random variable X if $h(\cdot)$ is a convex function.

- Thus if $\varepsilon_t \sim N(0, 1)$ $k_u \geq 3$.
- Even if $\varepsilon_t \sim N(0, 1)$, varying σ_t implies that R_t has non-normal distribution, with higher kurtosis.

Estimation of GARCH models

- GARCH cannot be estimated by ordinary least-squares. These models are estimated by *conditional maximum likelihood*.
- General Gaussian GARCH model

$$R_t = \mu_t(\phi) + \underbrace{\sigma_t(\phi, \psi)\varepsilon_t}_{u_t}$$

- Under the assumption that $\varepsilon_t \sim \text{i.i.d. } N(0, 1)$, and ϕ are the regression/ARMA parameters and ψ are the GARCH parameters.
- **Example:** AR(1)-GARCH(1,1) so

$$\mu_t = \phi_0 + \phi_1 R_{t-1}, \quad \sigma_t^2 = \omega + \alpha u_{t-1}^2 + \beta \sigma_{t-1}^2$$

hence $\phi = (\phi_0, \phi_1)'$ and $\psi = (\omega, \alpha, \beta)'$

- σ_t^2 depends on ϕ through $u_{t-1}^2 = (R_{t-1} - \mu_{t-1}(\phi))^2$.
- Conditional density $R_t|F_{t-1} \sim N(\mu_t, \sigma_t^2)$

$$f(R_t|F_{t-1}) = \frac{1}{\sqrt{2\pi\sigma_t^2(\phi, \psi)}} \exp\left(-\frac{1}{2} \frac{(R_t - \mu_t(\phi))^2}{\sigma_t^2(\phi, \psi)}\right),$$

Estimation of GARCH models

If we assume that $\mu_t(\phi)$ and $\sigma_t^2(\phi, \psi)$ depend on at most m lags of R_t, u_t and σ_t^2

Conditional joint density

$$f(R_T, \dots, R_{m+1} | F_m) = (\prod_{t=m+1}^T f(R_t | F_{t-1}))$$

- The conditional log-likelihood:

$$\begin{aligned}\mathcal{L}^*(\gamma | F_m) &= \sum_{t=m+1}^T \ell_t(\gamma) \\ \ell_t(\gamma) &= \log f(R_t | F_{t-1}), \\ \gamma &= (\phi', \psi')'\end{aligned}$$

Starting values for σ_t^2 and $u_t, t = 1, \dots, m$

$$\begin{aligned}u_t &= 0, \\ \sigma_t^2 &= \frac{1}{T-m} \sum_{t=m+1}^T (R_t - \mu_t(\phi))^2\end{aligned}$$

- Maximization of $\mathcal{L}^*(\gamma | F_m)$ can be done by numerical optimization algorithms.

Asymptotic Results

- Under general conditions (stationarity, existence of moments)

$$\sqrt{T}(\hat{\gamma}_{CML} - \gamma_0) \xrightarrow{D} N(0, A_0^{-1})$$

where $A_0 = E[-\frac{\partial^2 \ell_t(\gamma_0)}{\partial \gamma \partial \gamma'}]$ if the assumption of gaussianity holds (where γ_0) are the true values of the parameters.

- If we are not sure that $R_t|F_{t-1}$ are normally distributed and you use it anyway then we may still use the same estimation technique.
- This is called *quasi-maximum likelihood estimator (QML)*.
- QML is consistent under correct specification of both the conditional mean and the conditional variance.
- In this case

$$\begin{aligned} \sqrt{T}(\hat{\gamma}_{CML} - \gamma_0) &\xrightarrow{D} N(0, \mathcal{V}), \\ \mathcal{V} &= A_0^{-1} B_0 A_0^{-1} \end{aligned}$$

where $B_0 = E[\frac{\partial \ell_t(\gamma_0)}{\partial \gamma} \frac{\partial \ell_t(\gamma_0)}{\partial \gamma'}]$.

Asymptotic Results

- Standard errors of $\hat{\gamma}_{CML}$ can be obtained from the sandwich estimator for the asymptotic variance of $\hat{\theta}$:

$$\hat{\mathcal{V}} = \hat{A}_0^{-1} \hat{B}_0 \hat{A}_0^{-1} \xrightarrow{p} \mathcal{V}$$

where

$$\begin{aligned}\hat{A}_0 &= \frac{1}{T} \sum_{t=m+1}^T \left[-\frac{\partial^2 \ell_t(\hat{\gamma}_{CML})}{\partial \theta \partial \theta'} \right], \\ \hat{B}_0 &= \frac{1}{T} \sum_{t=m+1}^T \left[\frac{\partial \ell_t(\hat{\gamma}_{CML})}{\partial \theta} \frac{\partial \ell_t(\hat{\gamma}_{CML})}{\partial \theta'} \right]\end{aligned}$$

This in turn can be used to construct t -tests.

- The standard errors computed via this robust method are known (in this context) as *Bollerslev-Wooldridge standard errors*.

Estimation of GARCH models

Example:

Dependent Variable: Y

Method: ML - ARCH (Marquardt) - Normal distribution

Sample (adjusted): 12/03/1990 11/03/2005

Included observations: 3915 after adjustments

Convergence achieved after 15 iterations

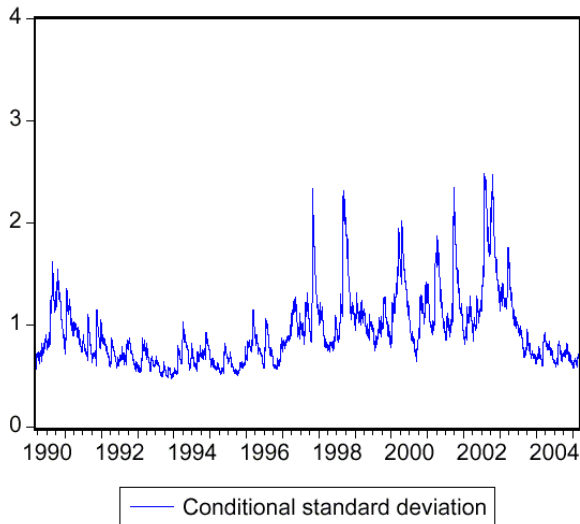
Bollerslev-Wooldrige robust standard errors & covariance

GARCH = C(2) + C(3)*RESID(-1)^2 + C(4)*GARCH(-1)

	Coefficient	Std. Error	z-Statistic	Prob.
C	0.053344	0.012855	4.149691	0.0000
Variance Equation				
C	0.008021	0.002392	3.353844	0.0008
RESID(-1)^2	0.058619	0.010235	5.727463	0.0000
GARCH(-1)	0.934423	0.009392	99.48822	0.0000

Estimation of GARCH models

- Fitted values for the variances



Specification Testing

Suppose you estimated an ARMA model and you would like to test if the model is well specified:

- Diagnostic tests are based on the *residuals* $\hat{u}_t$.
- If there are no ARCH effects we should find no autocorrelation in $\hat{u}_t^2$.
- Therefore, the model can be tested using Q -statistics for $\hat{u}_t^2$. The latter statistic tests if there are ARCH effects.
- Suppose that we would like to test that there are no ARCH effects.
- We can consider the model $E_{t-1}(u_t^2) = \gamma_0 + \gamma_1 u_{t-1}^2 + \dots + \gamma_m u_{t-m}^2$ and if there are no ARCH effects we must have $H_0 : \gamma_1 = \dots = \gamma_m = 0$
- Lagrange-Multiplier (LM) test against ARCH, which is obtained by $LM = T \cdot R^2 \xrightarrow{D} \chi^2(m)$ in the regression

$$\hat{u}_t^2 = \hat{\gamma}_0 + \hat{\gamma}_1 \hat{u}_{t-1}^2 + \dots + \hat{\gamma}_m \hat{u}_{t-m}^2 + e_t.$$

Specification Testing in GARCH models

Suppose now that you estimated a GARCH(q,s) model.

How do you test if the model is well specified?

- Diagnostic tests are based on the *standardized residuals* $\hat{\varepsilon}_t = \hat{u}_t / \hat{\sigma}_t$. If μ_t and σ_t are correctly specified, we should find no autocorrelation in $\hat{\varepsilon}_t$.
- We can apply the Q-statistics and Lagrange-Multiplier test for serial correlation in $\hat{\varepsilon}_t$.

How can we test if the model assumed for the conditional variance is well specified?

- We would like to test $H_0 : E_{t-1}(u_t^2) = \sigma_t^2$ which is equivalent to $E_{t-1}(\varepsilon_t^2 - 1) = 0$ with $\varepsilon_t = u_t / \sigma_t$.
- It is standard practice to apply the tests for ARCH effects described above, though they are *not valid* after the estimation of the GARCH model.

Specification Testing in GARCH models

A *valid* Lagrange-Multiplier (LM) test against ARCH is constructed in the following way:

- We can consider the model $E_{t-1}(\varepsilon_t^2) = \gamma_0 + \gamma_1 \varepsilon_{t-1}^2 + \dots + \gamma_m \varepsilon_{t-m}^2$ and if there are no further ARCH effects we must have $H_0 : \gamma_1 = \dots = \gamma_m = 0$.
- Lagrange-Multiplier (LM) test against ARCH, which is obtained by $LM = T \cdot R^2 \xrightarrow{D} \chi^2(m)$ in the regression

$$\hat{\varepsilon}_t^2 = \gamma_0 + \gamma_1 \hat{\varepsilon}_{t-1}^2 + \dots + \gamma_m \hat{\varepsilon}_{t-m}^2 + \delta' x_t + e_t.$$

where

$$x_t = \frac{1}{\hat{\sigma}_t^2} \frac{\partial \hat{\sigma}_t^2}{\partial \psi'},$$

ψ are the parameters of the conditional variance σ_t^2 and $\partial \hat{\sigma}_t^2 / \partial \psi'$ is the derivative of σ_t^2 with respect to ψ estimated under H_0 .

Volatility forecasting

- GARCH specification of

$$\begin{aligned}\sigma_h^2(1) &= \text{var}(u_{h+1}|F_h) \\ &= E(u_{h+1}^2|F_h)\end{aligned}$$

by construction gives one-step ahead forecasts of u_{h+1}^2 .

- Multi-step forecasts involves (using tower property) for $\ell \geq 1$

$$\begin{aligned}\sigma_h^2(\ell) &= \text{var}(u_{h+\ell}|F_h) \\ &= E(u_{h+\ell}^2|F_h) \\ &= E(E(u_{h+\ell}^2|F_{h+\ell-1})|F_h) \\ &= E(\sigma_{h+\ell}^2|F_h)\end{aligned}$$

For example for the GARCH(1,1) model we have

$$\begin{aligned}\sigma_h^2(\ell) &= E(\sigma_{h+\ell}^2|F_h) \\ &= E(\omega + \alpha u_{h+\ell-1}^2 + \beta \sigma_{h+\ell-1}^2|F_h) \\ &= \omega + (\alpha + \beta)\sigma_h^2(\ell - 1)\end{aligned}$$

- If $\alpha + \beta < 1$, where $\sigma^2 = \omega / (1 - \alpha - \beta)$

$$\sigma_h^2(\ell) = \sigma^2 + (\alpha + \beta)^{\ell-1}(\sigma_h^2(1) - \sigma^2) \rightarrow \sigma^2$$

as $\ell \rightarrow \infty$.

- If $\alpha + \beta = 1$ (IGARCH) then $\sigma_h^2(\ell) = \sigma_h^2(1) + \omega(\ell - 1)$